

JKBOSE 2026

CLASS 11TH

Anatomy Of Flowering Plants

LECTURE-02

BOTANY

ARSHID BHAT SIR



Topics To Covered

1

Tissue System (contd...)

2

Anatomy of Dicot & Monocot Root

3

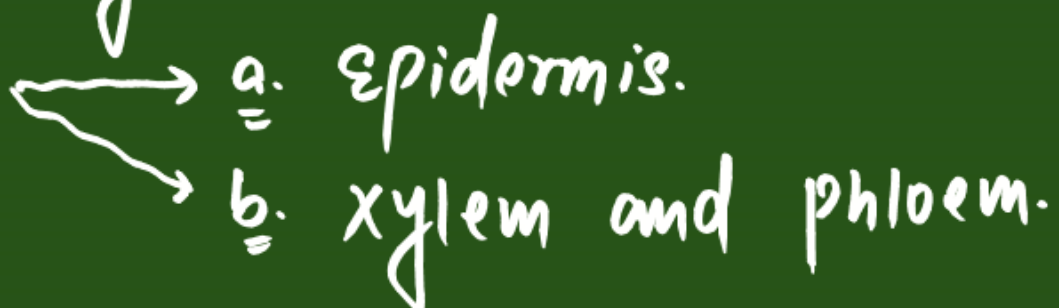
...

RECAP

→ 03 types of TISSUE SYSTEM:

- ① Epidermal tissue system. ✓
- ② Ground tissue system / Fundamental tissue system.
- ③ Vascular " " / Conducting " " .

GROUND TISSUE SYSTEM

- It is also called as fundamental tissue system.
- It consists of all the tissues except 
 - a. epidermis.
 - b. xylem and phloem.
- It is composed of simple permanent tissues like parenchyma, collenchyma, sclerenchyma.

GROUND TISSUE SYSTEM

Differentiated into

CORTEX

Pericycle

Medullary rays

Pith

Differentiated into

Hypodermis

General Cortex

Endodermis

NOTE

In leaves, ground tissue system is not differentiated and is called as Mesophyll.

VASCULAR TISSUE SYSTEM

- It is also called as Conducting tissue system because it transports water, H_2O , food.
- It consists of vascular bundle i.e., xylem and phloem.

VASCULAR TISSUE SYSTEM

TWO TYPES

Radial Vascular Bundle

Ex. Root

- xylem and phloem are present along different radii but alternating with each other.



Conjoint Vascular Bundle

- xylem and phloem are present along the same radii
- When xylem is present towards inner side and phloem towards outside, it is called as **Conjoint Collateral Vascular Bundle**.

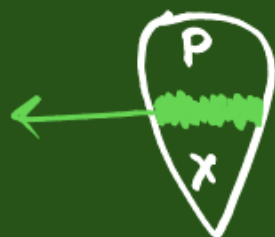


TWO TYPES

OPEN Conjoint Collateral vascular bundle (OR) OPEN V.B

- Meristematic tissue.
- Causes 2^o growth.

↑
Cambium



- Cambium present b/w xylem & phloem.
- Ex. Dicot Stem

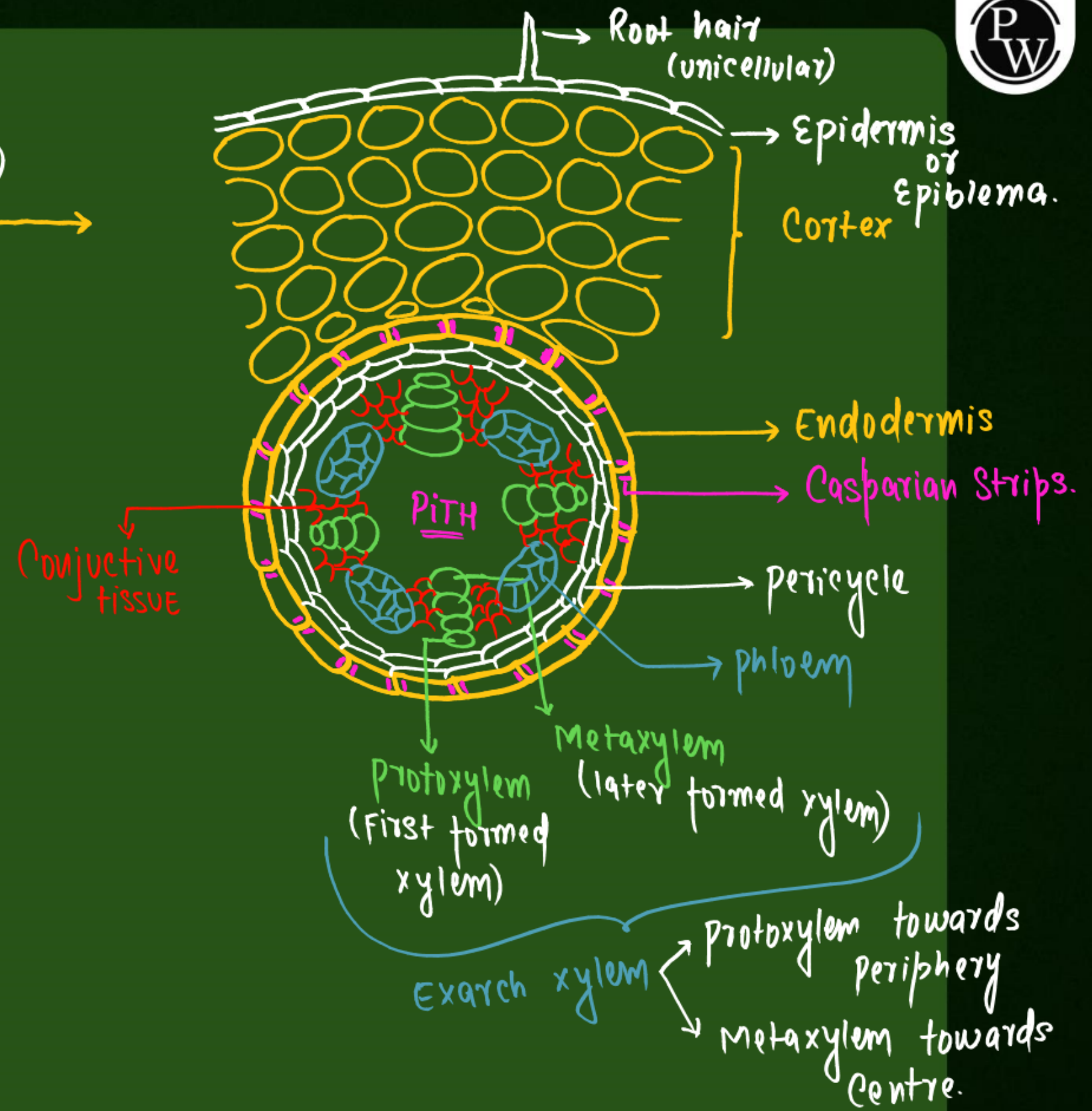
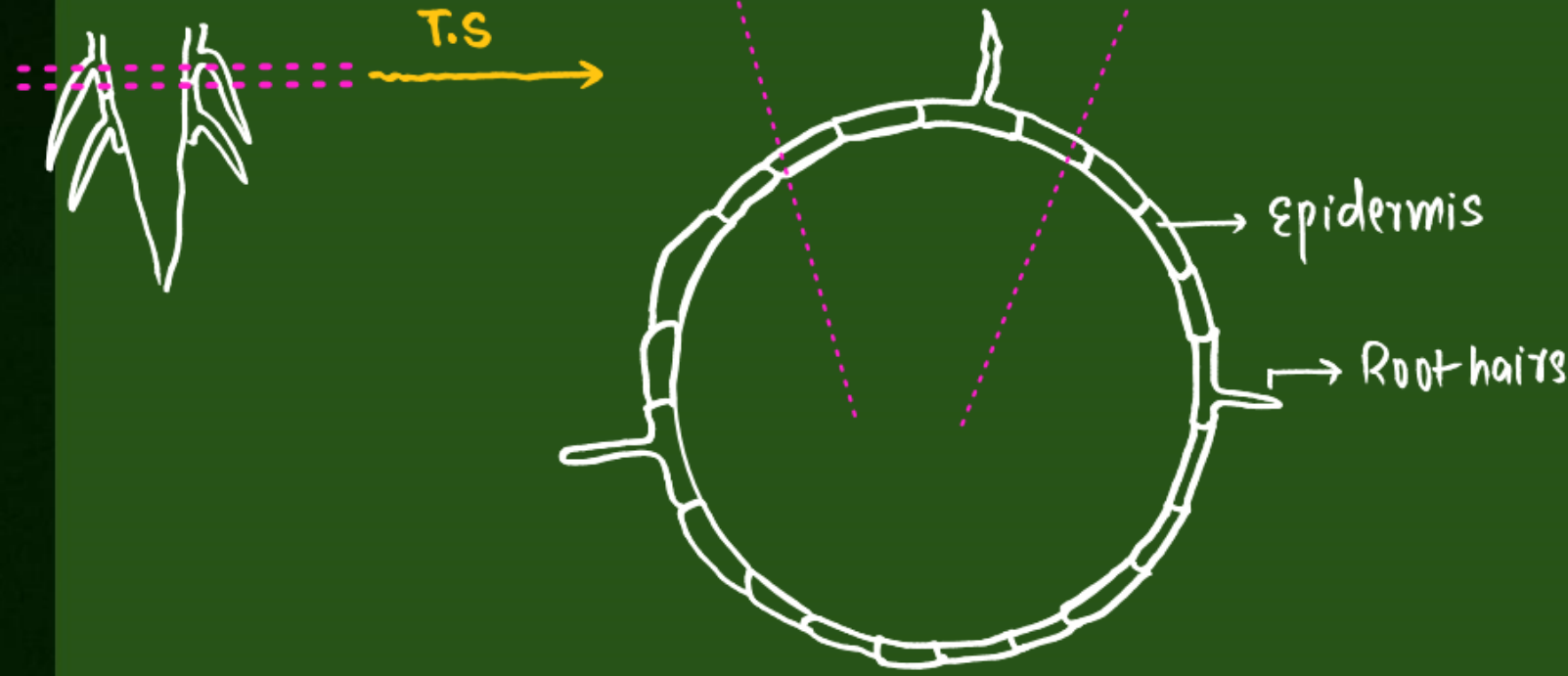
Closed Conjoint collateral V.B (OR) Closed V.B



- Cambium is -nt b/w xylem and phloem
- Ex. Monocot Stem, leaf.

ANATOMY OF DICOT ROOT

↳ Transverse Section of Root (maturation region)



EPIDERMIS: Outermost layer, single layered, possess unicellular root hairs.

CORTEX: Inner to epidermis, multilayered, parenchymatous (thin walled), intercellular spaces present.

ENDODERMIS: Innermost layer of cortex, single layered, barrel shaped, no intercellular spaces, radial and tangential walls possess Casparian strips (water impermeable waxy material made up of suberin).

PERICYCLE: Inner to endodermis, few layered, parenchymatous (thick walled), forms lateral roots and cambium during secondary growth.

VASCULAR BUNDLE: Radial, 2–4 xylem and phloem patches, xylem is exarch.

CONJUNCTIVE TISSUE: Parenchymatous, present between xylem and phloem.

PITH: small, inconspicuous (small).

ANATOMY OF MONOCOT ROOT

↓
NEXT lecture....

QUESTIONS

Closed vascular bundles lack:

(2012 Pre)

- A** Pith
- B** Ground tissue
- C** Conjunctive tissue
- D** Cambium

QUESTIONS

Ground tissue includes:

(2011 Pre)

- A** All tissues internal to endodermis
- B** All tissues external to endodermis
- C** All tissues except epidermis and vascular bundles
- D** Epidermis and cortex

Thank
You